

# Integration tests and fixtures

Project: Grounded Learning Service

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Testing model-dependent behaviour

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Least privilege and tool scopes

Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Typed contracts and schemas

Without opening the old lesson, explain one key idea from Typed contracts and schemas. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Trace the worked case for Integration tests and fixtures. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Integration tests and fixtures. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Apply a new case independently

Valid student lookup gets not-found from storage. Generator invents a capacity and final response appears supported. Locate failures.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

### My reasoning and evidence

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## 6. W2 Trace the worked case

A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file. Show the intermediate steps and give the conclusion with its scope.

### My reasoning and evidence

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## 7. W3 Audit the limit

A report omits this qualification: A mock verifies declared interactions, not real provider availability or all runtime limits. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

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## 8. W4 Apply a new case independently

Valid student lookup gets not-found from storage. Generator invents a capacity and final response appears supported. Locate failures.

**My reasoning and evidence**

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## 9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

### My reasoning and evidence

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## 10. Project checkpoint and reflection

Add a labelled integration tests and fixtures evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of integration tests and fixtures. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### My project change, supporting evidence and remaining limit

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## 11. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 12. Pictorial case practice: Integration tests and fixtures

Catch a wrong adapter connection. A service reads a fake catalogue fixture and answers a request for item N.

Fixture M="map", N="notebook". No external network is involved.

## Integration tests and fixtures

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Load fixture

The test supplies fixed records M and N.



#### 2. Follow identifier

The application must request N from its adapter.



#### 3. Check full reply

The returned item should be notebook with source N.

### CHANGE ONE THING

The adapter returns M when N was requested.

Explain the changed result using the matching lesson.

Follow request N through the fixture and adapter. Which source and item should appear in the reply, and what precisely fails when the adapter returns M?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

## 13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

### Evidence to keep · My record

**Evidence to keep:** This week's input and deciding step

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## 14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Catch a wrong adapter connection. A service reads a fake catalogue fixture and answers a request for item N.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? The returned item should be notebook with source N.

1. Load fixture
2. Follow identifier
3. Check full reply

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check full reply The returned item should be notebook with source N. Integration testing checks how components connect. Each component can work alone while their identifier exchange is wrong.

### Practice 2: explain the deciding evidence

Changed-case practice: The adapter returns M when N was requested. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Fail the integration check: map is the wrong item.

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Fail the integration check: map is the wrong item. The request now reaches evidence for another identifier.

**Repair a missing building block**

Review Testing model-dependent behaviour (ai-6/8). Goal: Explain the mechanism and terminology of testing model-dependent behaviour. Short practice: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?. Return to this week and explain what you changed.

Review Least privilege and tool scopes (ai-6/6). Goal: Explain the mechanism and terminology of least privilege and tool scopes. Short practice: Classify read S2, write S2 and read S5 as permitted or denied. Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.. Return to this week and explain what you changed.

Review Typed contracts and schemas (ai-6/3). Goal: Explain the mechanism and terminology of typed contracts and schemas. Short practice: Compare A's count 4 with B's count "4". Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?. Return to this week and explain what you changed.

**Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

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